

Shamus Li

Computational Imaging PhD Student

shamus@cs.cornell.edu

<https://shamus.li>

Education

PhD student in Computer Science, Cornell University 2024 – present

Advisor: Kristina Monakhova

MS in Electrical Engineering and Computer Sciences, Aug 2024
University of California, Berkeley

Advisor: Laura Waller

BS in Electrical Engineering and Computer Sciences, Aug 2023
University of California, Berkeley

Work Experience

Apple Sunnyvale, CA

Software Engineer Intern, Vision Products Group May 2024 – Aug 2024

- Created tooling that improved internal test automation coverage to near 100% and reduced quality-engineering test time by more than 5x.
- Collaborated with multiple cross-organizational teams to develop a novel task-driven camera simulation pipeline.

Lawrence Berkeley National Laboratory Berkeley, CA

Research Intern Oct 2022 – Aug 2023

- Built a human-in-the-loop semantic-segmentation pipeline for 3D cell volumes, reducing segmentation time from days to minutes.
- Designed an adaptive brush for organelle detection that improved efficiency by more than 10x compared with thresholding workflows.
- First undergraduate student to work at the National Center for X-Ray Tomography.

Meta Menlo Park, CA

VR Instructor Jun 2023 – Aug 2023

- Developed and taught a course on advanced VR experience design in partnership with Mission Bit.
- Worked with the State & Local Policy team to host events for students from underrepresented backgrounds in the Bay Area.

Palo Alto Research Center Palo Alto, CA

Research Intern Jun 2022 – Aug 2022

- Designed a multimodal-imaging workflow for building thermal analysis and an augmented-reality iOS app that improved sensor-commissioning setup time by 80%.

Teaching Experience

Cornell University

- Teaching assistant for Kristina Monakhova in Computational Imaging (CS 6662), Fall 2024 and Foundations of Computational Imaging (CS 4660), Spring 2026
- Teaching assistant for Bharath Hariharan and Wei-Chiu Ma in Computer Vision (CS 4670/5670), Spring 2025

University of California, Berkeley

- Teaching assistant for John DeNero and Pamela Fox in Structure of Interpretation of Computer Programs (CS61A), Fall 2021
- Teaching assistant for Nicholas Weaver in Great Ideas in Computer Architecture (CS61C), Fall 2022

Publications and Posters

Sparse Light Field Sampling Improves Casual 3D and 4D Reconstruction

Shamus Li, Ruiming Cao, Laura Waller, Kristina Monakhova, Sara Fridovich-Keil

ECCV 2026 S3DSGR Workshop

ICCP 2025 (Spotlight Poster)

When Do Diffusion Inverse Solvers Help in Lensless Imaging?

Shamus Li, John Thickstun, Kristina Monakhova

ICCP 2026 (Spotlight Poster)

Deep Optics Design for Extended-Field-of-View Depth-from-Defocus

Hasindu Kariyawasam, **Shamus Li**, Emma Alexander, Kristina Monakhova

ICCP 2026 (Spotlight Poster)

QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems

Cassandra Tong Ye, **Shamus Li**, Tyler King, Kristina Monakhova

arXiv preprint (2507.14760)

CVPR 2025 UNCV Workshop (Spotlight Presentation and Poster)

ICCP 2025, 2026 (Spotlight Posters)

INSTANT: 3D Interactive Semantic Segmentation of Cellular Structure in Soft X-Ray Tomograms

Shamus Li, Saman Mostafavi, Axel Ekman, Valentina Loconte, Sookyung Kim, Alejandro Brito, Carolyn Larabell

LBNL Molecular Biophysics & Integrated Bioimaging Workshop (Poster)

A Data-driven Method to Control Buildings Sub-zones to Spatio-temporal Profiles

Aayushman Sharma, Raman Goyal, Saman Mostafavi, **Shamus Li**, Alejandro Brito, Eric Bier
Building Simulation 2023

Sensorium: Commissioning Abundant Sensors with Augmented Reality and QR Codes

Eric Bier, Alejandro Brito, Raman Goyal, Saman Mostafavi, Lester Nelson, Ritesh Sharma, Mahabir Bhandari, Niraj Kunwar, **Shamus Li**

Building Simulation 2023

Honors

Outstanding Teaching Assistant Award, Cornell CS Department, Spring 2026

ICCP Summer School Fellow (Received scholarship to attend the summer school and the conference), 2025

Officer, Eta Kappa Nu (HKN) Honour Society, Mu Chapter

Dean's List, UC Berkeley EECS Department

Service

Reviewer for ECCV 2026 S3DSGR Workshop; IEEE Transactions on Computational Imaging, 2025; Nature Photonics, 2024.

Cornell University Computer Science Ph.D. admissions reader; 2025, 2026.

Cornell CS Student-Applicant Support Program mentor and Visit Days organizer; 2025, 2026.